MIL-A-37736
30 August 1978

MILITARY SPECIFICATION

ALUMINUM ACETATE AND ACETIC ACID SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Aluminum Acetate and Acetic Acid Solution (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 -Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be sterile Aluminum Acetate and Acetic Acid Solution suitable for otic administration.

3.1.1 Assays.

3.1.1.1 Aluminum oxide. The solution shall contain not less than 0.120 percent and not more than 0.145 percent (w/w) of aluminum oxide, when assayed as specified in 4.7.1.

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3.1.1.2 Total acetic acid. The solution shall contain not less than 2.25 percent and not more than 2.75 percent (w/w) total acetic acid when assayed as specified in 4.7.2.

3.1.1.3 Boric acid. The solution shall contain not more than 0.06 percent of boric acid when assayed as specified in 4.7.3.

3.1.2 Identification. The solution shall respond to the general identification tests for aluminum and for acetate in the USP.

3.1.3 pH. The pH of the solution shall be not less than 4.50 and not more than 6.00 at 25°C when determined potentiometrically using the USP method.

3.1.4 Specific gravity. The specific gravity of the solution at 25°C/25°C shall be not less than 0.99 and not more than 1.02 when determined using the USP method.

3.2 Sterility. The solution shall be sterile when tested in accordance with the USP.

3.3 Ingredients. All ingredients used in the manufacture of the solution shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quanity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.2.1 Sterility lot. A sterility lot is a lot or portion of a lot which (a) is sterilized in the same sterilizer load, (b) is filtered through a bacterial filter or filters into a retaining container, mixed and filled aseptically into final containers through a single filling apparatus or assembly in a period of time not to exceed one working day or shift, or (c) is subject to some other sterilizing process which shall be acceptable to the procuring activity. Upon approval of the procuring activity, the assigning of pharmaceutical lot (quality control) numbers without showing individual day's filling designation will be acceptable, provided samples from each day's filling are tested and reserve samples representing each day's filling are retained for future testing as needed.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

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4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assays	3.1.1	S-2	5
Identification	3.1.2	S-2	5
pH	3.1.3	S-2	5
Specific gravity	3.1.4	S-2	5
			Composite

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Non-conformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Solution not free from foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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4.7 Assays.

4.7.1 Aluminum oxide.

Pipet 25.0 ml of the sample solution into a tared 250 ml Erlenmeyer flask and weigh. Add, in order named and with continuous stirring, 25.0 ml of 0.05M disodium ethylenediaminetetraacetate and 20 ml of acetic acid-ammonium acetate buffer, TS, then heat the solution near boing point for 5 minutes. Cool, add 50 ml of alcohol, and 2 ml of dithizone, TS. Titrate with 0.05M zinc sulfate until the color changes from green-violet to rose-pink. Perform a blank determination, substituting purified water for the sample. Each ml of 0.05M Na_2EDTA is equivalent to 2.549 of Al_2O_3.

4.7.2 Acetic acid.

Accurately weigh about 4 grams of sample solution into a 50 ml beaker. Transfer to a 500 ml Kjeldahl flask with five 30 ml portions of purified water (150 ml total). Add 20 ml of phosphoric acid (85 percent) to the Kjeldahl flask. Connect the flask to a condenser, the delivery tube of which dips beneath the surface of 25 ml of 0.1N NaOH contained in a receiving vessel of at least 500 ml capacity. Distill, using low flame to prevent boiling over, until volume in Kjeldahl is reduced to about 50 ml. Remove flame, break connection between Kjeldahl flask and connecting tube, quickly add another 50 ml of purified water and reseal. Distill another 50 to 75 ml of liquid. When the distillation is complete, remove the receiving vessel from the apparatus, add 2 to 3 drops of phenolphthalein, TS to the contents and titrate the unreacted sodium hydroxide with 0.1N sulfuric acid to the disappearance of the pink color. Each ml of 0.1N NaOH is equivalent to 6.006 mg of $C_2H_4O_2$.

4.7.3 Boric acid.

Pipet 50.0 ml of sample solution into a tared 250 ml beaker and weigh. Transfer quantitatively into a 500 ml conical flask, rinsing beaker with 50 ml of purified water. Add 3 ml of phenolphthalein, TS, then add 0.1N sodium hydroxide from a buret until a faint pink color is obtained. Heat to boiling and again neutralize. Add 150 ml of glycerin to the neutralized solution and titrate with 0.1N NaOH. Perform a blank determination in the same manner, substituting purified water for the sample. Record the volumes of 0.1N NaOH used after the addition of the glycerin. Each ml of 0.1N sodium hydroxide is equivalent to 6.184 mg of H_3BO_3.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 60 ml, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available plastic squeeze bottle, with otic dropper tip, considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection require for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twenty-four intermediate packages (288 units of issue) shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four intermediate packages (288 units of issue) shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the puspose, so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-104-8061	ALUMINUM ACETATE AND ACETIC ACID SOLUTION, Otic, 60 ml

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6505-1885

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-37736 - ALUMINUM ACETATE AND ACETIC ACID SOLUTION

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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C. REASON FOR RECOMMENDED CHANGE(S)

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